

Meeting 1217

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12.17 Discussion

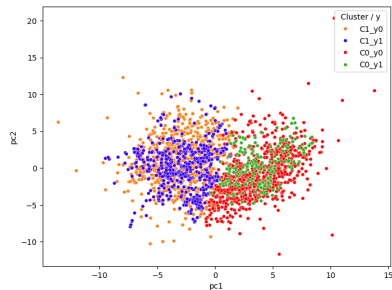
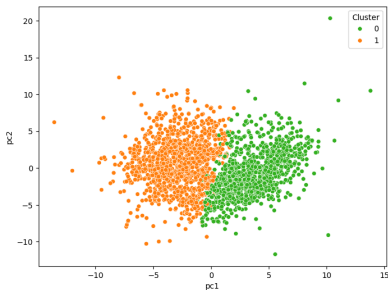
Semiconductor Manufacturing Process

- ▶ Given the fitted model $y = \hat{f}$
 - Describe what $\hat{f}$ is when x belongs to neighbors of X_0 for some “interesting” X_0
- ▶ 研究 $\hat{f}(x)$, $x \in \mathcal{N}(X_0)$, 觀察模型在 local 的解釋性、結構
 - 建構 $\hat{f}$ 的方法 (迴歸、機器學習), 使用鄰近資料 (y_k, x_k) 建立 local

12.17 Todo

- ▶ Do the SMOTE before PCA after delete outliers
- ▶ Decide a x_k (p dimension) from neighbors of X_0 to do the regression
 - 在每個 cluster 抽幾個代表點 X ，例如：
 1. cluster center (在 PCA 空間的重心，找最近的樣本)
 2. $y=1$ 且預測機率高的點
 3. 預測錯誤的點 (不穩定區)

Results I SMOTE + PCA + KMeans



cluster	0	1
$y = 0$	743	687
$y = 1$	205	510

Results I 5-folds cv - mean

model	ACC	FAR	ROCAU
LR	0.8520	0.1011	0.5459
SVM	0.9304	0.0041	0.5027
DT	0.8149	0.1415	0.5309
RF	0.9304	0.0048	0.5074
XGB	0.9272	0.0096	0.5145

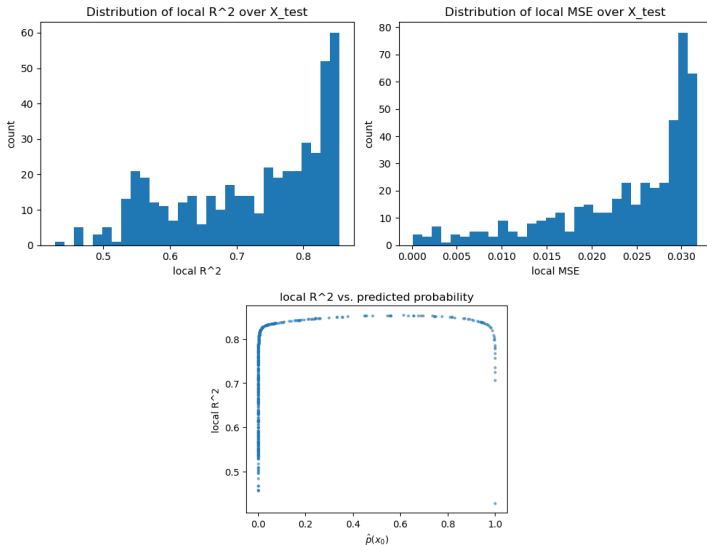
model	C ₀ ACC	C ₀ FAR	C ₀ ROCAU	C ₁ ACC	C ₁ FAR	C ₁ ROCAU
LR	*0.9219	*0.0942	*0.9431	*0.9131	0.1456	*0.9233
SVM	*0.9895	*0.0013	*0.9774	*0.9883	0.0044	*0.9870
DT	*0.9114	*0.0686	*0.8852	*0.8622	0.1716	*0.8681
RF	*0.9631	0.0000	*0.9146	*0.9749	*0.0014	*0.9708
XGB	*0.9799	0.0000	*0.9537	*0.9741	0.0248	*0.9739

Results II Build a surrogate model

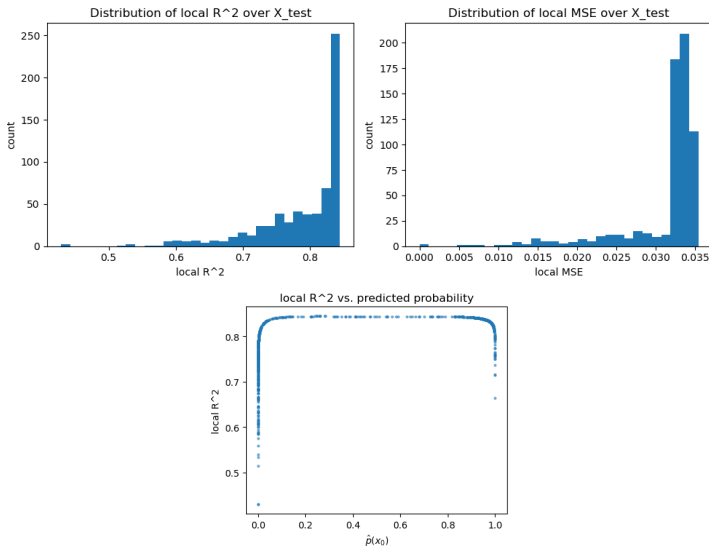
Reference: Ribeiro et al.(2016)

- ▶ Given a design point x_0
- ▶ Generate 擾動的 points $x_k = x_0 + \varepsilon_k$, where $\varepsilon_k \sim N(0, \sigma^2 I)$
- ▶ 代入 black-box model: $y_k = \hat{f}(x_k)$
- ▶ Fit a simple local model $g(x; x_0)$ on $\{(x_k, y_k)\}$ s.t.
 $\hat{f}(x) \approx g(x; x_0)$ near x_0 .

Results II No SMOTE



Results II SMOTE



The End

Questions & Comments